A data driven approach for optimal lineup building in Basket-Ball.

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Abstract

To Do

1 Introduction

Sports analytics has been a burgeoning field for decades. In team sports, data were primarily descriptive and high-level, focusing on direct measurements necessary to determine the outcome of a contest (number of points scored, win-loss records, etc.). The mid-20th century, through the professionalization of sports, saw the emergence of individual performance statistics, such as shots, tackles, and passes. These data, commonly referred to as **box-score** statistics, brought a player-centric dimension to performance analysis. Indeed, they introduced the possibility of attributing collective performance to individual players. Later development, with the increasing use of tracking systems in the 2010s, allowed to considerably enrich these data, adding more granularity. In the National Basketball Association (NBA), a real-time tracking technology, SportVU, was introduced into every arena in 2014. It allows differentiating shots no longer by their sole value (2 or 3 points) but by their location (elbow, post, etc.), or assess if a rebound is or is not contested.

While these statistics paved the way for aggregated indicators used to deepen the understanding of analyzed sports, with Sabermetrics in baseball (James, 1984) and Oliver's Four Factors in basketball (Oliver, 2004) as pioneering efforts, they remain widely used to assess individual performance. In the NBA, arguably the most competitive basketball league in the world, individual awards such as Most Valuable Player (MVP), Defensive Player of the Year (DPoY), selections to the NBA All-NBA Teams, and participation in the All-Star Game are to a great extent determined by players' statistical output. In short, box-score statistics are routinely used to identify and evaluate key players.

However, in team sports, while individual abilities and athletic level are crucial factors of performance, it is ultimately teams that win games and championships. Bridging the gap between individual performance analysis and team performance therefore represents a central challenge for sports analytics. More specifically, a fundamental question arises: how does the value of a team compare to the sum of the values of its individual constituents? And, conversely, how can one infer, from individual statistics alone, the value of a potential team? A seminal illustration of the use of box-score statistics to assemble a competitive team was presented in the book *Moneyball: The Art of Winning an Unfair Game*, later adapted into a major motion picture. This story describes how a small-budget baseball team managed to leverage Sabermetrics to build a competitive roster despite significant financial constraints. While *Moneyball* recounts a successful application of box-score-derived statistics, granting such data an overestimated relevance may yield opposite results. Since the 2010s, the NBA has witnessed several attempts at assembling so-called "Super Teams", few of which achieved sustained success. While no strict consensus on the term "Super Team" has been

established in the academic literature, NBA.com writer Steve Aschburner proposed the following defining characteristics[1]:

i It boasts two, three or more stars on its roster, any one of whom would be able to anchor a franchise on his own.

ii Trades and/or free agency had to have been used to unite those stars, either in their prime or at least after obvious, recognized success (All-Star appearances, all-NBA selections, major awards won).

iii The team needs to be a reasonable contender, not merely a collection of big names past their prime and far back of the reigning conference or league champs.

iv Bonus points if media outlets have carried one or more stories about the alleged Super Team wondering (a) if one basketball is enough, or (b) how the stars' strong personalities might gel or repel.

The recurring underperformance of such assemblies of individually dominant players relative to expectations highlights the limits of individual statistics as a sole proxy for team success, and motivates the development of more holistic analytical frameworks.

This work proposes a data-driven method, leveraging box-score statistics, to address the task of roster construction in basketball. Our methodology unfolds in two complementary steps:

1. We apply unsupervised learning to players' individual statistics in order to identify archetypes of players, thereby creating an intermediate level of analysis that transcends the athlete level, at which each individual is unique by definition. To this end, we make use of a dataset referenced in Muniz and Flamand (2022)[2]. We benchmarked several clustering algorithms to identify which performs best on this dataset.

2. We subsequently leverage these archetypes to **characterize each team according to the lineups it deploys**, and relate this characterization to team success. Lineups were extracted from play-by-play data, publicly available on Kaggle[3].

The remainder of this paper is organized as follows: Section 2 reviews related work on unsupervised learning applications in sports analytics and roster construction, Section 3 describes our methodology, Section 4 presents our results, which are discussed in Section 5, and Section 6 concludes the study.

2 Related Work

2.1 Unsupervised Learning in Sports

Machine Learning has been successfully applied for over a decade in sports analytics (Memmert, 2025). In this section, we are focusing on application of **unsupervised learning**. These approaches are primarily utilized to address the inherent lack of ground-truth labels in professional tracking data, focusing on player archetype discovery, tactical interaction modeling, and game-phase segmentation.

[1] https://www.nba.com/news/nba-super-teams-and-history, consulted on February 25th, 2026

[2] https://github.com/meganlynn12/jsa-cdkm?tab=readme-ov-file, consulted on February 25th, 2026

[3] https://www.kaggle.com/datasets/schmadam97/nba-playbyplay-data-20182019, consulted on February 25th, 2026

Discovery of Player Roles and Archetypes Unsupervised learning provides a robust mechanism for redefining player positions based on empirical performance rather than traditional categorical labels. Muniz and Flamand (2022) introduced a novel two-step methodology utilizing k-Means clustering to initialize similarity weights within a weighted network, followed by the Louvain method for community detection. Applied to NBA basketball tracking data (2014–2020), the study identified eight distinct archetypes, such as "Superstar Guards" and "Role Playing Bigs", which explained significantly more variation in team success (Win Shares) than the traditional five-position model. While this study relates closely to our archetypes optimization objective, if differs as clusters are different across seasons, when we need comparable clusters to consistently characterize teams' lineups. In soccer, Bialkowski et al. (2014) utilized minimum entropy data partitioning, an Expectation-Maximization (EM) approach to detect team formation in presence of player permutations. By learning roles directly from spatiotemporal data, they found that a dynamic role-based representation allows for accurate formation detection and provides necessary context for analyzing individual player behavior that is lost in static identity models.

Tactical Interactions Miller and Bornn (2017) implemented a hierarchical model involving probabilistic clustering via Bezier curves and Latent Dirichlet Allocation (LDA) to discover an offensive "vocabulary" in the NBA. This unsupervised approach successfully grouped similar possessions, such as "weave" plays or specific tactical sequences leading to corner three-pointers, enabling efficient search and exploration of vast tracking databases. Knauf et al. (2015) developed a novel class of Spatio-Temporal Convolution Kernels (STCKs) applied through kernelized k-medoids and spectral clustering. Utilizing Bundesliga soccer data, they demonstrated that these kernels effectively identify recurring movement patterns such as "game initiations" and "scoring opportunities". Their findings highlight that STCKs are capable of distinguishing between multifaceted team philosophies, such as the rehearsed short moves of one team versus the linear, long-ball strategies of another, more effectively than baseline Hausdorff distance or bag-of-words methods. In handball, Schrapf et al. (2017) employed self-organizing maps (specifically the DyCoN neural network) to cluster offensive and defensive constellations. They found that specific defensive patterns were significantly more successful (measured via odds ratios) against certain offensive behaviors, providing an objective framework for preparation against opponents. Leser et al. (2019) applied spectral clustering following Laplacian Score-based feature reduction to analyze youth soccer 3-vs-2 games. This unsupervised mining identified "hidden" tactical strategies, revealing that offensive efficiency is maximized when attacks are initiated via the wings and concluded with a backward "through-ball" pass against the playing direction. Fewell et al. (2012) treated NBA teams as dynamic networks, using clustering coefficients and network entropy to quantify offensive strategies. They found that team-level success was positively correlated with higher network entropy (unpredictability) and lower degree centrality, suggesting that elite teams tend to distribute decision-making across multiple players rather than relying on a single dominant ball-handler.

Game Phase Segmentation Perše et al. (2009) utilized Gaussian Mixture Models (GMM) and the EM algorithm to segment basketball trajectories into game phases (offense, defense, and timeouts) with approximately 90% accuracy. Furthermore, they used k-means clustering to partition the court into semantic regions, enabling the automatic recognition of organized play types using Levenstein distance between observed actions and database templates. Montoliu et al. (2015) proposed a Bag-of-Words (BoW) strategy to recognize activities in soccer without requiring individual player tracking. By extracting "motion words" from optical data, they achieved high classification accuracy (up to 92.8%) for team activities such as Ball Possession, Quick Attack, and Set Pieces.

2.2 Roster Optimization

The application of machine learning and mathematical optimization has become a cornerstone of modern sports analytics, shifting the focus from individual talent to the synergistic effects of player combinations. In soccer, Abidin (2021) implemented several machine learning algorithms to evaluate players based on training data and coach assessments. The study demonstrated that Random Forest models could select optimal player combinations with 93.9% reliability, producing lineup suggestions that were 97.2% similar to a professional coach's ideal team. In Basketball, Siddique et al. (2025) adopted a two steps methods to (i) predict team wins based on players' statistics and (ii) optimize roster composition. They suggest three formulations of a linear programming optimization problem, the first aiming at maximizing the win rate of a hypothetical team, the second minimizing the salary expense given the constraint of a minimum win rate to achieve and the last aiming at maximizing the team's shooting coverage. Penner (2025) moved toward positionless basketball by using K-means clustering to identify nine distinct player archetypes across 110 global leagues, subsequently utilizing multiple linear regression to rank athletes by "points per minute" to optimize championship roster construction. In the specialized context of wheelchair basketball, Calvo et al. (2025) combined Bayesian longitudinal modeling with stochastic integer linear programming to identify optimal lineups while strictly adhering to the Player Classification Points System (PCPS) constraints. This methodology allows coaches to incorporate uncertainty into the selection process, effectively estimating the probability that a five-player configuration will yield the highest team performance. Beyond traditional field sports, Sapienza et al. (2019) explored team composition in the Multiplayer Online Battle Arena games Dota 2, employing deep neural autoencoders to recommend teammates who maximize an individual's "TrueSkill" score gain. Their model achieved a 19.5% improvement in recommendation accuracy over traditional graph factorization methods. Such approaches are supported by theoretical frameworks like those proposed in Liemhetcharat and Veloso (2012), which utilize synergy graphs and simulated annealing to model how task-based distances between heterogeneous agents affect overall team capability.

3 Methodology

3.1 Data

We relied on three data sources in this work: (i) a tracking-improved box-score dataset for identifying players' archetypes (see paragraph 3.1.1), (ii) play-by-play data to retrieve lineups used by teams and their relative +/- statistics (see paragraph 3.1.2) and (iii) team performance metrics to assess success of teams (see paragraph 3.1.3). We only retain in our analysis complete seasons where these three sources are available, and considered only regular season games. Playoff games were discarded, as we believe they could bring a distortion to the analysis. Indeed, only the best performing teams of a season qualify to the playoffs, facing stronger adversity, which could deteriorate their performance. Moreover, this choice allows to have exactly the same number of games for all NBA teams, and therefore not over represent good performing teams. Our dataset covers 5 NBA seasons, from 2015/16 to 2019/20, representing a total of 5,980 games (1,230 per season from 2015/16 to 2018/19 and 1,060 for the 2019/20 season, due to the modification of the season's schedule incurred by the Covid-19 pandemic). An overview of the datasets can be found in Table 1.

3.1.1 Tracking-improved box-score data

The dataset presented in this paragraph has been referenced in Muniz and Flamand (2022) and is publicly available on their github repository[4]. It consists in data obtained from the website

[4]Consulted on February 25th, 2026

Table 1: Datasets overview

Season	Nb games	Nb players*	Nb Play-by-play lines	Nb lineups
2015-16	1,230	472	601,557	12,759
2016-17	1,230	484	596,645	12,960
2017-18	1,230	532	590,868	13,869
2018-19	1,230	508	614,516	14,761
2019-20	1,060	523	539,265	13,811
Total	**5,980**	**2,519**	**2,942,851**	**68,160**

* Notes: (i) the total number of player does not correspond to unique players as players can play several seasons, (ii) we discarded players that were missing in one or more datasets. None of them play more than 10 minutes per game on average.

Table 2: Tracking-improved dataset description

Dataset	Description	Raw Source(s)
Scoring	2/3 FG(M/A/%)	Shooting
	(Drive/Catch Shoot/Pull Up/Paint/Post/Elbow) Touch PTS and %	Tracking Efficiency
	PTS per (elbow/post/paint) touch	Tracking Possessions
	PTS (off TOV/2nd chance/fastbreak/in the paint)	General Misc
	USG%, Off. Rating	General Traditional
	DD2, TD3	General Traditional
Passing/ Playmaking	AST %, AST:TO, AST Ratio	General Advanced
	Time of possession, touches (all & front-court), avg. sec & dribble per touch	Tracking Possessions
	Drive passes, AST, TOV	Tracking Drives
	AST, FT AST, Secondary AST, Potential AST, AST:Pass	Tracking Passing
	Pass/AST/TOV per (elbow, post, paint) touch	Tracking Elbow, Post, Paint Touch
	Screen AST	Hustle
Rebounding	OREB/DREB (Contest/Uncontest/Chance/Distance/Defer)	Tracking Rebounding
	Off/Def Box-outs, Box-out REBs	Hustle
Defense	STL, BLK, Def Rim FGM/A	Tracking Defense
	Def FGM/A, % difference (FG%)	Tracking Defensive Impact
	Opp 2nd chance/fastbreak/in-paint points	Tracking Defense
	Deflections, Charges Drawn, Contested 2/3 PT shots	Hustle
Hustle/ Miscellaneous	Loose balls recovered	Hustle
	Personal fouls drawn	General Traditional
	Drive personal fouls	Tracking Drives
	Distance traveled, average speed	Tracking Speed Distance

Adapted from Muniz and Flamand (2022). For an extensive description of the statistics, please refer to https://www.nba.com/stats/help/glossary

nba.com, initially split in 21 raw datasets and reorganized by the authors of Muniz and Flamand (2022) in 6 datasets as described in Table 2. Please note that contrarily to Muniz and Flamand, we did not consider the "Clutch" dataset as it contains data from particular part of games (less than 5 points difference in the last 5 minutes), and therefore overlaps with other datasets. These data contains traditional basketball box-score statistics (e.g. shot attempted, shot made, rebound, assists, etc.) augmented from Sports VU tracking system (introduced in NBA arenas in 2014). Tracking improved box-score statistics as it (i) brought higher granularity to existing indicators (for instance, categorizing shots by areas on the pitch rather than sole value, identifying if a rebound or a shot is contested or not, etc.) and (ii) created new indicators (by recording for instance the number of boxouts[5]).

In these datasets, every entry corresponds to a player's average statistics over a season. We concatenated seasons and datasets and removed of duplicated columns (minutes and number of games played are recorded in each dataset). In total, the concatenated dataset contains 2,519

[5]A boxout is a situation where a player manages to position himself between the basket and his opponent in a shooting situation, to achieve higher probability of securing the rebound.

entries (player's season) distributed into 30 NBA teams, and has 129 dimensions (performance indicators). We manually excluded features being collinear by design (e.g. TS_{PCT}, the True Shooting Percentage, is obtained from other shooting statistics, AST and TOV, the total number of assists and turnovers is the sum of the number of assists and turnovers by locations, etc.), and we discarded some ratio features, which reduced the number of features to 74. We subsequently (i) transformed statistics expressed in volumes in per 48 minutes statistics so players can be compared, (ii) scaled our dataset using a minmax scaler and (iii) further reduced the dimensionality of our dataset by applying a Principal Component Analysis (PCA) and retaining 80% of explained variance.

3.1.2 Play-by-play data

The dataset presented in this section is publicly available Kaggle[6]. Play-by-play data record timestamp-indexed of event in a game. Recorded event encompasses traditional box-scores events (e.g. shot attempts, rebounds, assists, etc.) and the identity of the players credited for the corresponding events, as well as game-related information (start of period, end of period, player substitutions, etc.). From the identity of players credited for event and the changes log, we were able to identify for all timestamps the composition of both team participating in a game. A composition (also referred to as lineup) is a combination of 5 players from the pool of available players. In the NBA, teams are allowed to register 12 players for a given game, corresponding to $\binom{12}{5} = 792$ possible combinations. In practice, teams often have more than 12 players in their roster, and the selection of the 12 registered for a game is already a one-out-of-many choices, and teams can also sign players for short-duration contracts, trade them in the course of a season, etc. Therefore, the figure of 792 possible combinations is no theoretical maximum of the number of lineups. In total, we identified between 12,759 and 14.761 unique lineups per season. Teams aligned between 187 and 794 different lineups over the course of a season ($\mu = 454.4, \sigma = 114.7$). For each lineup, we computed (i) the time it has played and (ii) its +/- statistics. The +/- statistic corresponds to the difference of points vs opponents when the analyzed lineup is on the floor, a positive value indicating the lineup outscores its opponent and a negative value the opposite. It is a well-used indicator on how much difference makes a player or a combination of players.

3.1.3 Team performance data

We collected from nba.com two indicators of teams' success:

i Teams' Win Rate (number of games won / number of games played), ranging from 0.122 to 0.890 ($\mu = 0.500, \sigma = 0.152$).

ii A binary indicator worth 1 if the considered team reached the conference semi-finals in the playoffs and 0 otherwise. The NBA is split into two conferences (East and West), both having their own playoff competition. The NBA champion is the winner of the best-of-seven series between East and West champions. Reaching the conference semi-finals is therefore equivalent to being within the last 8 contenders for the NBA title, which corresponds to the top 26.7%.

An illustration of the distribution of success can be found in Figure 1. Top performing teams appear to be the Golden State Warriors 2015, 2016, 2017 and the San Antonio Spurs 2016, achieving win rates above 80% and reaching Conference semi-finals, while Minnesota Timberwolves 2015 and Philadelphia 76ers 2016 emerge as low performers with win rate below 20%. Atlanta Hawks 2017 and 2019, and Memphis Grizzlies 2017 and 2018 aligned over 700 different lineups during the season, and Minnesota Timberwolves 2017, Orlando Magic 2018 and Detroit Pistons 2015 less than 240. None of these teams reached the Conference semi-finals.

[6]Consulted on February 25th, 2026

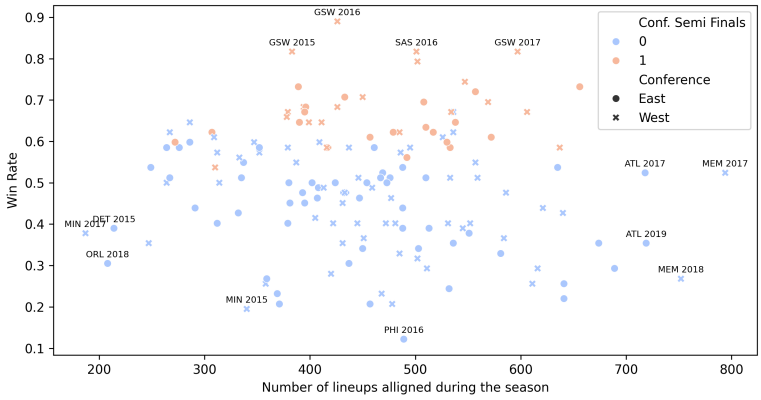

Figure 1: Distribution of teams according to their number of lineups aligned during each season (x-axis) and win rate (y-axis). Red dots indicate teams reaching the Conference semi-finals, and blue dots the opposite, Crosses correspond to Western Conference teams and dots to Eastern Conference teams.

3.2 Clustering of NBA players

From the tracking-improved boxscore dataset (see 3.1.1), we aim at identifying players' archetypes. To do so, we need to create a partition of our dataset. We benchmarked various approaches, including centroid-based, hierachical agglomerative, density-based and gaussian models, with a hyperparameters search. We implemented these approaches with the scikit-learn Python library (Pedregosa et al., 2011).

Among partition-based approaches, the centroid-based k-means algorithm partitions observations into k groups by minimizing within-cluster dispersion, typically defined as the sum of squared Euclidean distances between observations and their corresponding cluster centroids. Hierarchical clustering methods provide an alternative framework by constructing nested partitions without directly optimizing a global objective. In particular, agglomerative hierarchical clustering proceeds iteratively by merging pairs of clusters according to a linkage criterion (single, complete, average, or Ward's minimum-variance linkage). The resulting dendrogram encodes a multi-scale representation of the data, allowing cluster structure to be examined at different levels of granularity. Density-based approaches define clusters as connected regions of high sample density. HDBSCAN extends the DBSCAN framework by building a hierarchy based on mutual reachability distances and extracting clusters according to their stability across density levels. This formulation enables the identification of clusters with varying densities while treating low-density regions as noise, which is particularly relevant in high-dimensional and irregularly distributed datasets. However, in our case, we allocated "noisy" players to the nearest cluster (based on distance to cluster's centroid), as (i) we see no reason to discard outliers and (ii) there is a significant risk high performing (and therefore highly represented) players would be considered as outliers. Finally, model-based clustering relies on probabilistic generative assumptions. Gaussian Mixture Models (GMMs) assume that observations arise from a finite mixture of Gaussian distributions, each parameterized by a mean vector and covariance matrix. Model parameters are typically estimated via the EM algorithm, yielding soft probabilistic assignments of observations to latent components.

Details about the methods, hyperparameters searched and hyperparameters values can be found in Table 3.

Comparison metric We compared different clustering algorithms to determine which yields the best results on our dataset. Contrary to supervised learning, where a form of consensus emerges around some key metrics (e.g. Accuracy, F1 score and ROC-AUC for classification tasks, RMSE

Table 3: Algorithms and Hyperparameters tested

Algorithm	Hyperparameters	Range
K-Means	Number of clusters Type	{2, ..., 20} Standard, Spherical
Agglomerative	Number of clusters Linkage function	{2, ..., 20} Ward, Average, Complete
HDBSCAN	Minimum Cluster Size Minimum Samples Cluster Selection Epsilon Maximum Cluster Size	{4, 6, 8, 10, 15, 20, 30, 40} {4, 6, 10, 20} {0, 0.2, 0.5, 1, 2} {None, 400, 800}
GMMs	Covariance Type Number of components	Spherical, Diagonal, Full, Tied {2, ..., 20}

and its variants for regression tasks), Clustering Validity Indices (CVIs) are still subject to discussion within the academic community. Arbelaitz et al. (2013) compared 30 CVIs and concluded three groups of indices emerge, and the first of them contains indices that "behave better", with "the differences being statistically significant". Well know metrics such as Silhouette Score, Davies-Bouldin, Calinski-Harabasz, etc. are listed as part of this well performing group. On the other hand, Gagolewski et al. (2021) stated it is better "not to treat them as objective functions for identifying good partitions", and specified this is particularly the case for Silhouette and Davies-Boulin. We therefore chose to rely on the Calinski-Harabasz index to compare different methods. This metrics is defined as follows:

$$CH = \frac{n-k}{k-1} \times \frac{\sum_1^n \|\mu - \mu_{C(i)}\|^2}{\sum_1^n \|x_i - \mu_{C(i)}\|^2} \tag{1}$$

where n is the population size, k the number of clusters, $C(i)$ is the clusters in which point $x_i, i \in \{1, ..., n\}$ is allocated, μ and $\mu_{C(i)}$ are the centroids of the population and of cluster $C(i)$ respectively, and $\|a - b\|$ represents the Euclidian distance between points a and b.

Once an optimal method is identified, we investigated several criterions to identify the best number of clusters:

i Clustering method specific indicators (e.g. K-Means inertia)

ii Gap Statistic (Tibshirani et al., 2001), which compares the total within intra-cluster variation for different number of clusters with their expected values under null reference distribution of the data. The estimate of the optimal clusters will be value maximizing the gap statistic (equivalently, yielding the largest gap statistic), corresponding to the clustering structure the farthest away from the random uniform distribution of points. Algorithm 1 summarizes this approach.

3.3 Lineups analysis

From the lineups identified (see Section 3.1.2), we retain for further analysis only those aligned more than 24 minutes (which corresponds to half the duration of a NBA game). Applying this filter corresponds to keeping 4,266 lineups out of the 68,160 lineups identified (6.3%), representing 51.3% of playtime. A brief description of lineups and their usage in terms of playtime can be found in Table 4. We computed for all lineups (i) the time they have played and (ii) their +/- statistics. We subsequently computed the average +/- per 48 minutes. From the partition of NBA players performed (see Section 3.2), we characterize each lineup by its composition in terms of clusters. Each lineup can be expressed as $L_C = \{k : \text{count}(C(p_i) = k), i \in L_P \forall k \in \{1, ..., n_k\}\}$ where

Algorithm 1 Gap Statistic (from Tibshirani et al., 2001)

Input: Data matrix $\{x_{ij}\}$, $i = 1, \ldots, n$, $j = 1, \ldots, p$;
Maximum number of clusters $k_{\max}$;
Number of reference data sets B.
Output: Optimal number of clusters $\hat{k}$.
$\hat{k} \leftarrow None$
for $k = 1, \ldots, k_{\max}$ **do**
  while $\hat{k} = None$ **do**
    Cluster the observed data and compute the within-cluster dispersion:
    $W_k = \sum_{r=1}^{k} \frac{1}{2n_r} D_r$ where $D_r = \sum_{a,b' \in C_r} \|a - b\|$
    for $b = 1, \ldots, B$ **do**
      Generate a reference data set X^* from a uniform null distribution.
      Cluster X^* and compute W^*_{kb}.
    end for
    Compute the Gap statistic and its corrected standard deviation:

$$\mathrm{Gap}(k) = \frac{1}{B} \sum_{b=1}^{B} \log(W^*_{kb}) - \log(W_k), \qquad s_k = \sqrt{1 + \frac{1}{B}} \cdot \mathrm{sd}(k)$$

    if $\mathrm{Gap}(k) \geq \mathrm{Gap}(k+1) - s_{k+1}$ **then**
      $\hat{k} = k$
    end if
  end while
end for

Table 4: Lineup distribution

Time Bracket	Time (minutes)	Count
0 - 12 minutes	191,313	58,494
12 - 24 minutes	89,904	5,400
24 - 36 minutes	50,574	1,735
36 - 48 minutes	33,449	809
1 - 2 games	70,686	1,071
2 - 3 games	33,071	286
> 3 games	108,741	365

L_C (respectively L_P) is the lineups characterized in terms of clusters (respectively players), $C(p_i)$ indicates the clusters in which belongs player p_i and n_k the number of clusters of the selected partition.

First, for all clusters $C_i, i \in \{1, ..., n_k\}$ taken individually, we computed the percentage of time in which at least one player from C_i is playing. We then extracted L_{C_k} and $L_{\bar{C}_k}$, the +/- per 48 minutes of the ensemble of lineups containing at least one player from cluster C_k and the ensemble of lineups without any player from cluster C_k Then, we perform a non-parametric Mann-Whitney U-test on L_{C_k} and $L_{\bar{C}_k}$ to evaluate the impact of a cluster's presence or absence on the +/- per 48 minutes statistic. A peut être supprimer si on l'inclut dans un cadre plus global To be completed ?

Our objective is to find the best compositions in terms of clusters, L^*_C. We note $S(L)$ the performance of lineup L and define the performance of a given lineup as its +/- per 48 minutes. We want L^*_C to be the solution of Equation 2 given constraint presented in Equation 3.

$$S(L^*_C) = \max_{L \in \Omega_L} S(L) \tag{2}$$

$$L = \{k : c_k, k \in \{1, ..., n_k\}\} \quad \text{with} \quad \sum_{k=1}^{n_k} c_k = 5 \tag{3}$$

where Ω_L is the universe of possible compositions and c_k represents the number of players of cluster k present in the lineup.

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We decompose the performance of a lineup as follows:

$$S(L) = \sum_{i=1}^{5} \sum_{\{P_1,...P_i\} \in L} S(\bigcap(C(P_1), ...C(P_i))) \tag{4}$$

where $C : \Omega_P \mapsto \Omega_C$ is the function associating player P to the cluster c it belongs to, $\bigcap(C(P_1), ...C(P_i))$ represents the possible combinations up to i players P belonging to lineup L. Ω_P and Ω_C represents respectively the universe of players and clusters. For instance, $\bigcap(C(P_1))$ represents the performance associated with the presence of cluster $C(P_1)$ in a lineup and $\bigcap(C(P_1), C(P_4), C(P_5))$ represents the performance associated with the combined presence of clusters $C(P_1)$, $C(P_4)$ and $C(P_5)$.

For each lineup L and for $i \in \{1, ..., 5\}$, we assessed $S(\bigcap(C(P_1), ...C(P_i))), \{P_1, ...P_i\} \in L$ as follows:

i We split our dataset D in two: D_w where the condition (combined presence of $(C(P_1), ...C(P_i))$) is met and $D_{\bar{w}}$ where it is not. We note n the number of observations in D_w

ii We create a biased distribution of D_w and $D_{\bar{w}}$ where all observation are weighted by time spent playing,

iii We compute $\bar{D_w}$ and s, the mean and standard deviation of D_w and μ the mean of $D_{\bar{w}}$

iv We compute Student's T-statistic, $T = \frac{\bar{D_w} - \mu}{s/\sqrt{n}}$ and perform a Student T test with a significance level of $\alpha = 0.05$.

v If the test is positive (p-value $< \alpha$), we set $S(\bigcap(C(P_1), ...C(P_i))) = \bar{D_w} - \mu$. Else we set $S(\bigcap(C(P_1), ...C(P_i))) = 0$

In practice, we determine a threshold $n_0 = \min(30, N * 0.1)$ below which steps (ii) to (v) are not performed, and $S(\bigcap(C(P_1), ...C(P_i))) = 0$. To mitigate the impact of such limitation, and maintain comparability over lineups, we limit the summation of Equation 4 according to the share of lineups not reaching the threshold of n_0.

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4 Results

4.1 Clustering

4.1.1 Principal Component Analysis

As specified in Section 3.1.1, we performed a PCA to reduce the dimensionality of our dataset. We set an explained variance threshold target of 80%, which corresponds to 14 principal components. Please refer to Table 5 for an overview of principal components, and to Table 8 in appendix for detailed coordinates of Principal Components in feature space.

4.1.2 Clustering algorithms

The Calinski-Harabasz indices of the compared method are presented in Figure 2. Please note that the number of clusters of for some algorithms (e.g. K-Means, GMM) an input hyperparameter, but for some others (e.g. Partition - HDBSCAN), it is a consequence of other hyperparameters. A clear hierarchy emerges between algorithm : K-Means, Spherical K-Means and GMM consistently achieve top performance, while Agglomerative with Complete and Average linkage emerge as worst performers. Agglomerative with Ward linkage consistently falls below the top three performers and Partition HDBSCAN present more variations with cluster sizes, performing at the

Table 5: Principal Component details

PC	EV%*	Play creation	Defense	Shooting	Rebound	Other	Description
1	0.240	**Dribble** (↑) AST (↑)	2FG contest (↓)	Post % (↓), 3FGA (↑), Own shoot (↑)	**OREB contest** (↓) Overall (↓)	Poss (↑)	Perimeter & ball handler
2	0.144	Touch (↑) AST (↑)	n.s.	**Post % (↑), Elbow %** (↑) Paint % (↑), Drive % (↑)	OREB contest (↑)	**Minutes** (↑)	High usage inside player
3	0.088	Poss (↓)	2FG contest (↓)	**Elbow %** (↑), Paint % (↑), Post % (↑), 3FG% (↑)	n.s.	Minutes (↑)	Efficient shooter with low poss
4	0.058	Post touch and AST (↑)	(↓)	**Post % (↑), Elbow % (↓)** Paint % (↓)	Def. boxouts(↓)	Minutes (↑) Usage % (↑)	Post play & poor defense
5	0.045	Post touch (↓), Elbow touch (↓)	**2/3 FG contest** (↑) Deflections (↑)	Post FG% (↑), 3FGA (↑), 3FG% (↑)	**Def. boxouts(↑)**	Screen AST (↑)	Inside defense
6	0.040	Post touch (↓)	2FG contest (↓)	**Post %** (↑), Elbow % (↑)	n.s.	Minutes (↓)	Inside shoot
7	0.031	Post & elbow touch (↑)	Deflection (↓)	**Elbow % (↑), Paint %** (↑)	OREB contest (↓)	n.s.	Low hustle shoot
8	0.029	Elbow pass (↑)	Deflection (↓)	**Drive FG%** (↑), FG2% (↑)	OREB contest (↓)	Usage % (↓)	Shot creation
9	0.027	Passes (↑)	Deflection (↓)	**Post %** (↑), Elbow % (↓)	OREB contest (↓)	Minutes (↓)	Catch & shoot
10	0.024	Passes (↑)	Def rim % (↑)	Paint % (↓), Drive % (↑)	**OREB contest** (↑)	n.s.	Drive & hustle
11	0.023	Paint AST (↑)	Deflection (↑)	3FGA (↓), 2FGA (↓)	n.s.	**Minutes** (↑)	High min., low resp. def.
12	0.019	Post touch (↑)	**Overall** (↑)	**Pull-up %** (↑)	n.s.	Usage % (↓)	Polyvalent defense
13	0.018	Post AST (↓)	**Def rim %** (↑)	Overall (↓)	Def. boxouts(↑)	Minutes (↑)	Rim defense
14	0.018	Post AST (↑)	**Def rim %** (↑)	Drive % (↑), **overall** (↓)	Def. boxouts(↑)	n.s.	In. def. & shot creation

↑ (respectively ↓) indicates feature positively (resp. negatively) correlated with the corresponding PC.
EV% = Explained Variance Ratio. Features in bold indicate the most relevant for the corresponding PC.

same level as GMM for the number of clusters for which it ranks best, and as bad as Agglomerative with complete linkage for the others. From this analysis, we will use K-Means as partition algorithm for the remaining of the paper.

For determining the optimal number of clusters, we relied on (i) an elbow method based on the value of K-Means inertia defined as

$$Inertia = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_{C_i}\|^2 \tag{5}$$

where μ_{C_i} represents the centroid of the clusters containing x, and k the number of clusters, and (ii) the gap statistic method (Tibshirani et al., 2001). For the latter, we ran 100 iterations of Algorithm 1 with $B = 10$. Figure 3 displays the results for both approaches. From the elbow method, the optimal number of clusters is between 6 and 13 and from the gapstat algorithms, it appears between 9 and 11. We therefore chose 10 clusters for the selected partition, as 10 is the mode of our gapstat optimal number of clusters and falls within the Elbow method range.

4.1.3 Selected partition

Figure 4 presents a graphical display of the selected partition, and Table 6 a description of the clusters as well as notable players. For an extensive clusters' centroid position in Principal Component space (respectively, feature space), please refer to Table 9 (respectively Table 10) in appendix. From Table 6, we can detect two clusters of "superstar" players: cluster 5 with star guards / small forwards, with high shooting statistics and playmaking, high usage rate, but relatively low rebounding and defensive impact, and cluster 10 with two-ways impacting "bigs" (Power Forward - PF and Centers - C), presenting the highest playmaking characteristics among clusters containing PFs and Cs, good rebounding and defensive impact (even if player 4 and 9 outperform them in the latter). The remaining clusters contain players who could be regarded as "supporting cast". Cluster 1 players display high shooting percentages and touch the ball less often. They play a rather high number of minutes, and participate in the rebounding tasks. Cluster 2 contains players with predilection shooting position (Elbow and 3 pointers), with low ball handling and playmaking statistics. Cluster 3 players participate in the game creation with ball handling, assists, and passes,

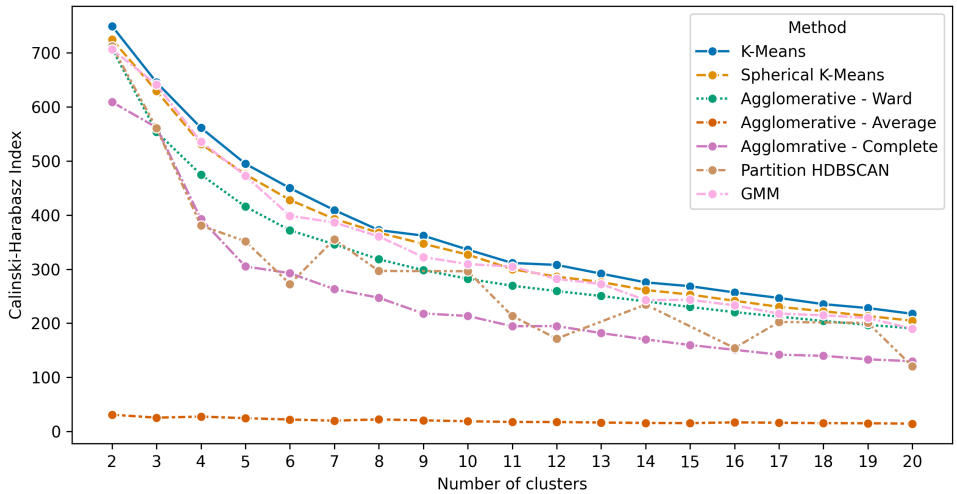

Figure 2: Comparison of clustering algorithm.

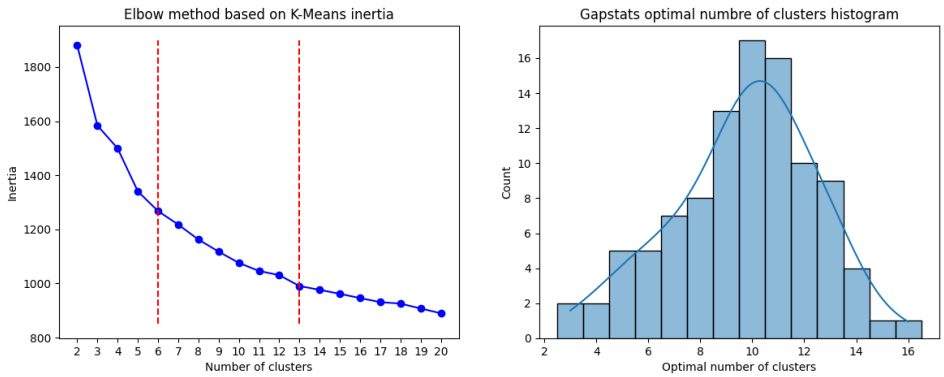

Figure 3: Determination of the optimal number of clusters. The left panel displays the inertia value of K-Means and its evolution and the right panel a histogram of the best number of clusters determined from the gap statistic method, over 100 repetitions.

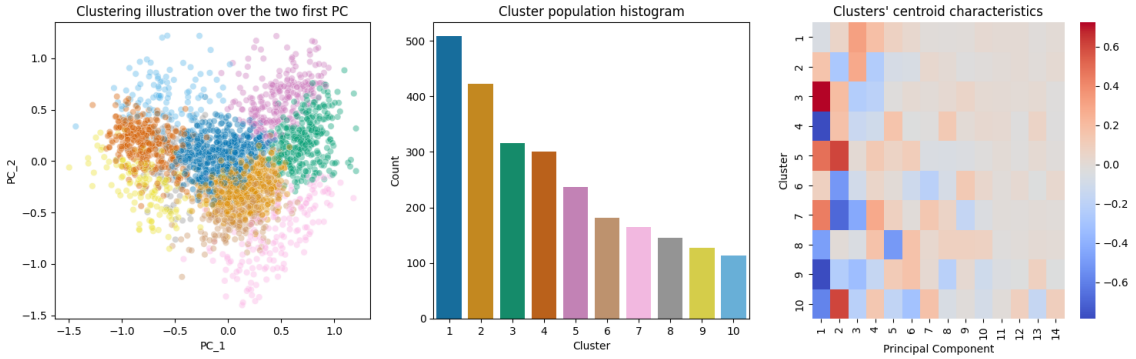

Figure 4: Visualization of the selected partition. The left panel represents players in a 2D space determined by the two first principal components, the middle panel a histogram of cluster populations and the right panel clusters' centroid positions along the principal components space.

Table 6: Clusters presentation

Cluster	Description	Positions*	Notable players
1	Polyvalent shooter with few possession	SF, PF	Khris Middleton, Andrew Wiggins
2	3pts shooter, low playmaking	SG, SF	Avery Bradley, Gary Harris
3	Playmaker with low rebounding	PG	Kyle Lowry, C.J. McCollum
4	Rim protector, rebounder, pick & roll player	C	Rudy Gobert, Clint Capela
5	High usage rate playmaker	PG, SG	James Harden, LeBron James
6	Low usage, low playmaking 3pt shooter	SG, SF, PF	Iman Shumpert, Malik Monk
7	Ball carrier with low FG%	SG, PG	Mike Miller, Wade Baldwin IV
8	Inside shooter with high rebounding	C, PF	Joffrey Lauvergne, Lavoy Allen
9	Low scoring, eff. screen, rebound and defense	C, PF	Tyson Chandler, Ed Davis
10	High offensive impact bigs	C, PF	Anthony Davis, Karl Anthony Towns

Note *: The "position" columns contains the most represented listed positions of players present in the cluster, ordered by frequence. PG: Point Guard, SG: Shooting Guard, SF: Small Forward, PF: Power Forward, C: Center.

display good own-shoot (drive and pull-ups) capacity, but little rebounding.* Players of cluster 4 present very high rebounding and defense statistics, participate in play development with screens and touches / assists from the paint, and display rather good shooting statistics from specific location (post and paint). Cluster 6 players shoot from the 3 point range, are average in terms of defense and rebounding, and low in terms of ball possession and playmaking. Players from cluster 7 dribble more than the average player, display low shooting, defensive and rebounding statistics. Cluster 8 contains big men with good shooting capacity from the post, the paint and the elbow, but bad percentages on own-shoot creation. They participate in playmaking with assists from the post and the elbow, and display good rebounding capacity. Compared to cluster 8's, players from cluster 9 are big men with higher defensive impact, who participate in the play development with their screens. They also display good rebounding statistics.

4.2 Lineups analysis

From the clustering of players and the subsequent characterization of lineups, we identified 459 unique lineups (from 4,266 when players were considered instead of clusters). Figure 5 displays the +/- per 48 minutes statistic depending on the presence or absence of clusters, as well as

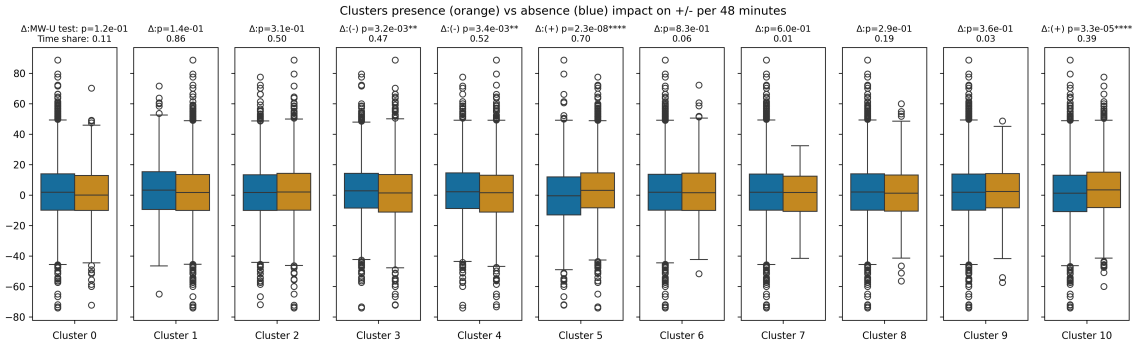

Figure 5: Boxplots of +/- per 48 minutes statistics decomposed by clusters' presence (orange) or absence (blue). p-values correspond to Mann-Whitney U-test, with the following significance levels: * = p<0.05, ** = p<0.001, *** = p<0.0001, **** = p<0.0001. (+) indicates significant over-performance and (-) indicates significant under-performance. Time shares indicate the percentage of time with at least one player of the corresponding cluster playing.

Table 7: Lineup combinations

Combination size	n_L	o/w valid	o/w significant	o/w valid and significant	Valid rate
1	10	10	5	5	1.00
2	55	50	17	16	0.91
3	220	126	42	35	0.57
4	715	114	32	21	0.16
5	2002	17	8	0	0.01

We considered a lineup as 'valid' where the sample size is greater than n_0 and significant where its t-test p-value is below 0.05.

the Mann-Whitney U-test p-values. We observe a disparity in terms of clusters' use: Clusters 1 and 5 are highly represented, appearing in respectively 86% and 70% of playtime. On the other hand, Clusters 7 and 9 only appear 1% and 3% of the time. For most 6 out of 10 clusters, no statistically significant differences in +/- per 48 minutes can be found between lineups containing and lineups without players from the analyzed cluster. However, we note the following exceptions:

i Lineups with superstar players (both guard and bigs, clusters 5 and 10) significantly outperforms lineups without these superstars,

ii Lineups with players from cluster 3 (playmakers who display low rebounding abilities) and 4 (rim protectors) significantly underperform.

To be completed

Table 7 displays the number of possible combinations of clusters and the number of valid possibilities in our dataset (where a candidate lineup is considered as valid when $n \geq n_0$, n being the sample size associated with the lineup and n_0 the threshold defined in Section 3.3)

4.3 Roster Constitution

To do

5 Discussion

To do

Limitations : cluster stability, financial consideration, few seasons, bootstrapping issue (non explored space) Fewell et al. (2012)

6 Conclusion

To do

References

Abidin, D. (2021). A case study on player selection and team formation in football with machine learning. *Turkish J. Electr. Eng. Comput. Sci.*, 29:1672–1691.

Arbelaitz, O., Gurrutxaga, I., Muguerza, J., Pérez, J. M., and Perona, I. (2013). An extensive comparative study of cluster validity indices. *Pattern Recognit.*, 46:243–256.

Bialkowski, A., Lucey, P., Carr, P., Yue, Y., Sridharan, S., and Matthews, I. (2014). Large-scale analysis of soccer matches using spatiotemporal tracking data. In *2014 IEEE international conference on data mining*, pages 725–730. IEEE.

Calvo, G., Armero, C., Grimm, B., and Ley, C. (2025). Selecting the Best Compositions of a Wheelchair Basketball Team: A Data-Driven Approach. *The American Statistician*, 79(2):212–220.

Fewell, J. H., Armbruster, D., Ingraham, J., Petersen, A., and Waters, J. S. (2012). Basketball teams as strategic networks. *PloS one*, 7(11):e47445.

Gagolewski, M., Bartoszuk, M., and Cena, A. (2021). Are cluster validity measures (in) valid? *Information Sciences*, 581:620–636.

James, B. (1984). The bill james baseball abstract.

Knauf, K., Memmert, D., and Brefeld, U. (2015). Spatio-temporal convolution kernels. *Machine Learning*, 102:247 – 273.

Leser, R., Hoch, T., Tan, X., Moser, B., Kellermayr, G., and Baca, A. (2019). Finding efficient strategies in 3-versus-2 small-sided games of youth soccer players. *Kinesiology*.

Liemhetcharat, S. and Veloso, M. M. (2012). Modeling and learning synergy for team formation with heterogeneous agents. In *Adaptive Agents and Multi-Agent Systems*.

Memmert, D. (2025). *Artificial Intelligence and Machine Learning in Sports Science*. Springer Verlag GmbH, DE.

Miller, A. C. and Bornn, L. (2017). Possession sketches: Mapping nba strategies. In *Proceedings of the 2017 MIT Sloan Sports Analytics Conference*, volume 1. IEEE.

Montoliu, R., Martin-Felez, R., Torres-Sospedra, J., and Martinez-Uso, A. (2015). Team activity recognition in association football using a bag-of-words-based method. *Human movement science*, 41:165–78.

Muniz, M. and Flamand, T. (2022). A weighted network clustering approach in the NBA. *Journal of Sports Analytics*, 8(4):251–275.

Oliver, D. (2004). *Basketball on paper: rules and tools for performance analysis*. U of Nebraska Press.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Penner, L. S. (2025). Player archetypes within basketball: optimizing roster composition to create a championship team. *Frontiers in Sports and Active Living*, 7:1639431.

Perše, M., Kristan, M., Kovačič, S., Vučkovič, G., and Perš, J. (2009). A trajectory-based analysis of coordinated team activity in a basketball game. *Computer Vision and Image Understanding*, 113(5):612–621.

Sapienza, A., Goyal, P., and Ferrara, E. (2019). Deep neural networks for optimal team composition. *Frontiers in big Data*, 2:14.

Schrapf, N., Alsaied, S., and Tilp, M. (2017). Tactical interaction of offensive and defensive teams in team handball analysed by artificial neural networks. *Mathematical and Computer Modelling of Dynamical Systems*, 23:363 – 371.

Siddique, S., Li, L., and Wang, Y. (2025). Finding optimal teams: An analysis of nba statistics and constraints. *2025 IEEE 11th International Conference on Big Data Computing Service and Machine Learning Applications (BigDataService)*, pages 26–34.

Tibshirani, R., Walther, G., and Hastie, T. (2001). Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63.

Appendices

Table 8: Comprehensive description of Principal Components in features space

Feature	PC 1	PC 2	PC 3	PC 4	PC 5	PC 6	PC 7	PC 8	PC 9	PC 10	PC 11	PC 12	PC 13	PC 14
EFG PCT	-0.032	0.053	0.062	-0.037	0.026	-0.053	-0.008	0.069	0.092	-0.020	-0.002	-0.074	0.059	-0.097
USG PCT	0.048	0.179	-0.038	0.082	-0.068	-0.186	-0.025	-0.154	-0.119	-0.141	-0.308	-0.103	0.035	-0.025
MIN	0.060	0.351	0.252	0.112	-0.087	-0.309	-0.072	-0.109	-0.183	-0.184	0.589	-0.041	0.251	0.031
PTS OFF TOV	0.013	0.032	0.014	0.007	-0.004	-0.036	-0.013	-0.022	-0.015	-0.037	-0.047	-0.035	0.010	-0.031
PTS 2ND CHANCE	-0.112	0.058	-0.025	-0.011	-0.045	-0.050	-0.031	-0.056	0.010	-0.080	-0.113	-0.080	0.096	-0.120
PTS FB	0.023	0.021	0.024	-0.003	-0.012	-0.012	-0.042	-0.016	-0.034	-0.029	-0.034	-0.027	0.018	-0.035
PTS PAINT	-0.094	0.144	-0.080	-0.040	-0.065	-0.019	-0.155	0.010	-0.100	-0.151	-0.171	-0.114	0.060	-0.096
DRIVE PTS	0.122	0.126	-0.021	-0.005	0.004	-0.047	-0.160	0.011	-0.155	-0.037	-0.144	-0.091	0.012	-0.022
DRIVE FG PCT	0.019	0.175	0.082	-0.043	0.014	-0.236	-0.275	0.700	-0.151	0.252	-0.063	-0.215	-0.255	0.244
CATCH SHOOT PTS	0.034	-0.018	0.144	0.057	0.053	-0.191	0.177	-0.080	0.120	0.073	-0.053	-0.111	-0.074	-0.006
CATCH SHOOT FG PCT	0.017	0.088	0.141	0.014	0.013	-0.168	0.160	-0.012	0.215	0.113	0.047	-0.229	0.092	-0.188
PULL UP PTS	0.129	0.096	0.021	0.042	0.013	-0.101	-0.023	-0.024	-0.028	-0.075	-0.148	0.071	0.111	-0.073
PULL UP FG PCT	0.073	0.117	0.129	0.049	-0.007	-0.222	0.052	0.291	0.131	0.101	-0.191	0.668	0.186	-0.415
PAINT TOUCH PTS	-0.213	0.067	-0.093	-0.046	-0.059	0.013	-0.074	0.006	0.038	-0.137	-0.075	-0.075	0.100	-0.098
PAINT TOUCH FG PCT	-0.095	0.182	0.262	-0.304	-0.037	0.050	-0.397	-0.136	0.626	-0.291	-0.055	0.066	-0.168	0.176
POST TOUCH PTS	-0.110	0.107	-0.048	0.126	-0.107	-0.135	0.110	-0.064	-0.031	-0.128	-0.134	0.091	-0.147	0.069
POST TOUCH FG PCT	-0.217	0.363	0.261	0.621	0.378	0.457	-0.009	0.041	0.019	0.009	-0.076	0.011	-0.043	0.029
ELBOW TOUCH PTS	-0.097	0.059	-0.021	-0.042	-0.058	-0.005	0.077	0.034	-0.025	-0.029	-0.065	-0.040	0.047	-0.045
ELBOW TOUCH FG PCT	-0.049	0.297	0.450	-0.539	-0.104	0.305	0.422	-0.013	-0.317	0.046	-0.115	-0.002	-0.048	-0.005
FG2A	-0.034	0.133	-0.081	0.027	-0.093	-0.056	-0.077	-0.042	-0.106	-0.123	-0.189	-0.033	0.062	-0.068
FG2 PCT	-0.083	0.091	0.066	-0.108	-0.017	0.017	-0.151	0.187	0.099	-0.073	-0.037	-0.039	0.039	-0.097
FG3A	0.182	-0.050	0.197	0.070	0.191	-0.287	0.148	-0.229	0.057	0.044	-0.195	-0.133	-0.134	0.112
FG3 PCT	0.121	0.051	0.200	0.035	0.108	-0.254	0.101	-0.045	0.208	0.178	-0.010	-0.141	-0.058	-0.152
OFF LOSSES RECOV.	-0.012	0.028	-0.024	-0.089	0.174	-0.072	-0.077	-0.028	-0.106	0.008	-0.069	0.020	0.054	0.145
DEF LOSSES RECOV.	0.009	0.002	-0.010	-0.037	0.078	-0.024	-0.013	-0.020	-0.046	-0.015	0.001	0.029	-0.008	0.057
PFD	-0.022	0.119	-0.063	0.018	-0.043	-0.067	-0.064	-0.042	-0.101	-0.087	-0.088	-0.047	0.062	-0.021
DRIVE PF	0.080	0.079	-0.018	0.012	-0.019	-0.025	-0.114	-0.050	-0.127	-0.037	-0.102	-0.076	0.027	-0.025
AVG SPEED OFF	0.059	-0.031	0.004	-0.080	0.068	0.088	-0.055	0.027	0.075	0.104	-0.049	-0.060	0.028	-0.116
AVG SPEED DEF	0.005	-0.061	0.010	-0.047	0.007	0.071	-0.038	0.043	0.001	0.007	-0.004	0.026	0.010	-0.046
STL	0.026	0.014	-0.026	-0.030	0.014	0.039	-0.059	-0.037	-0.073	-0.033	0.092	0.124	-0.118	0.003
BLK	-0.109	0.024	-0.039	-0.018	-0.009	-0.009	-0.019	-0.010	-0.004	-0.001	0.002	-0.044	-0.054	-0.128
DEF RIM FGM	-0.069	0.006	-0.039	-0.018	0.027	-0.016	0.020	-0.001	0.003	0.035	-0.062	0.030	0.073	0.081
DEF RIM FGA	-0.088	0.009	-0.046	-0.015	0.020	-0.019	0.021	0.007	0.006	0.025	-0.041	-0.002	0.020	0.015
DEF RIM FG PCT	0.086	0.027	0.051	-0.044	0.038	-0.009	-0.021	-0.031	0.039	0.110	-0.132	0.274	0.445	0.542
D FGM	-0.050	-0.031	-0.073	-0.019	0.033	0.022	0.019	-0.020	0.009	0.013	-0.118	0.016	0.015	0.056
D FGA	-0.052	-0.035	-0.071	-0.019	0.032	0.023	0.014	-0.014	0.003	0.002	-0.100	0.007	-0.047	-0.019
D FG PCT	-0.008	-0.002	-0.020	-0.002	0.010	0.001	0.022	-0.016	0.007	0.027	-0.053	0.037	0.126	0.161
OPP PTS FB	-0.003	0.007	-0.003	0.001	-0.017	0.008	0.013	0.002	-0.010	-0.006	-0.033	-0.006	0.040	0.027
OPP PTS PAINT	0.006	0.004	-0.001	-0.018	0.051	-0.027	-0.009	-0.018	-0.031	0.002	-0.061	0.030	0.107	0.107
DEFLECTIONS	0.022	0.029	-0.036	-0.144	0.279	-0.051	-0.145	-0.085	-0.219	-0.082	0.187	0.341	-0.325	-0.036
CHARGES DRAWN	0.007	0.026	-0.001	-0.014	0.068	-0.056	0.006	-0.025	-0.037	-0.001	0.072	0.034	-0.021	0.016
CONTESTED SHOTS 2PT	-0.210	0.032	-0.113	-0.172	0.404	-0.143	0.016	-0.044	-0.061	-0.058	-0.009	0.001	-0.096	-0.155
CONTESTED SHOTS 3PT	-0.006	-0.023	0.001	-0.075	0.229	-0.073	-0.055	-0.063	-0.116	-0.046	-0.010	0.116	-0.164	-0.054
SCREEN ASSISTS	-0.182	0.047	-0.119	-0.109	0.220	-0.061	0.063	0.036	0.028	-0.060	0.075	-0.041	0.129	-0.061
SCREEN AST PTS	-0.201	0.051	-0.131	-0.121	0.248	-0.066	0.069	0.038	0.030	-0.066	0.087	-0.049	0.140	-0.067
PAINT TOUCH PASSES	-0.114	0.025	-0.073	-0.019	-0.059	0.052	0.003	0.045	0.029	0.030	0.073	0.006	0.030	-0.043
PAINT TOUCH AST	-0.080	0.093	-0.040	0.007	-0.043	-0.047	0.000	0.039	0.004	-0.028	0.224	0.028	0.034	0.017
PAINT TOUCHES	-0.129	0.031	-0.073	-0.015	-0.048	0.022	-0.024	0.012	-0.003	-0.038	-0.013	-0.031	0.056	-0.058
POST TOUCH PASSES	-0.083	0.076	-0.061	0.062	-0.096	-0.102	0.115	-0.026	-0.008	-0.052	-0.029	0.106	-0.162	0.104
POST TOUCH AST	-0.100	0.117	-0.081	0.081	-0.139	-0.154	0.156	-0.033	0.015	-0.063	0.000	0.169	-0.288	0.201
POST TOUCHES	-0.120	0.105	-0.076	0.107	-0.143	-0.151	0.146	-0.062	-0.030	-0.106	-0.104	0.126	-0.197	0.103
ELBOW TOUCH PASSES	-0.160	0.067	-0.110	0.006	-0.125	-0.012	0.133	0.116	0.101	0.030	0.096	0.033	0.034	0.078
ELBOW TOUCH AST	-0.088	0.055	-0.071	-0.008	-0.056	-0.037	0.099	0.056	0.077	0.006	0.072	0.029	0.021	0.073
ELBOW TOUCHES	-0.225	0.101	-0.135	-0.003	-0.157	-0.029	0.151	0.125	0.065	0.007	0.031	0.006	0.072	0.039
AST	0.173	0.210	-0.169	-0.045	0.052	0.037	0.031	-0.019	0.070	0.038	0.111	0.008	-0.062	0.061
FT AST	0.023	0.022	-0.021	0.005	-0.016	0.020	0.005	-0.011	0.008	-0.004	0.013	-0.007	-0.011	-0.011
SECONDARY AST	0.067	0.059	-0.018	-0.016	0.009	0.006	0.002	-0.020	0.015	0.007	0.025	-0.003	0.005	-0.010
POTENTIAL AST	0.200	0.223	-0.198	-0.039	0.013	0.093	0.027	-0.029	0.079	0.030	0.121	0.019	-0.084	0.019
PASSES MADE	0.085	0.212	-0.240	-0.070	0.052	0.096	0.143	0.011	0.202	0.228	0.103	-0.004	-0.056	0.002
PASSES RECEIVED	0.226	0.248	-0.209	-0.029	0.020	0.039	0.078	-0.038	0.099	0.058	-0.046	-0.024	-0.018	-0.016
DRIVE PASSES	0.113	0.071	-0.052	-0.032	0.023	0.035	-0.052	-0.023	-0.023	-0.008	-0.023	-0.027	0.013	-0.014
DRIVE AST	0.062	0.042	-0.031	-0.014	0.011	0.016	-0.023	-0.008	-0.002	0.009	0.001	-0.030	-0.002	-0.014
TOUCHES	0.101	0.254	-0.233	-0.036	0.026	0.021	0.107	-0.051	0.128	0.148	-0.032	-0.047	-0.042	-0.013
AVG DRIB PER TOUCH	0.350	0.178	-0.180	-0.051	0.043	0.144	-0.117	-0.004	-0.057	-0.061	-0.075	0.000	0.092	-0.096
TOV	0.026	0.091	-0.102	-0.000	-0.017	0.001	-0.014	-0.054	-0.048	-0.030	-0.095	-0.042	-0.018	0.031
OREB UNCONTEST	-0.044	0.003	-0.036	-0.017	0.002	0.024	0.001	0.045	0.015	-0.069	0.005	-0.035	0.036	-0.047
OREB CONTEST PCT	-0.284	0.127	0.022	-0.030	-0.173	-0.043	-0.375	-0.416	-0.107	0.675	0.001	0.087	-0.005	-0.083
OREB CHANCE DEFER	-0.021	0.061	0.005	0.007	-0.027	-0.024	-0.028	-0.017	-0.048	-0.020	0.102	-0.017	0.079	-0.047
DREB CONTEST	-0.100	0.027	-0.048	0.002	-0.034	-0.024	0.034	-0.015	0.018	0.023	-0.027	-0.043	-0.032	-0.009
DREB UNCONTEST	-0.076	0.052	-0.048	0.001	0.001	-0.042	-0.007	-0.043	-0.019	0.014	-0.010	-0.047	0.034	0.035
DREB CONTEST PCT	-0.144	0.011	-0.027	0.013	-0.067	-0.022	0.052	-0.002	0.030	0.089	-0.042	-0.051	-0.113	-0.040
DREB CHANCE DEFER	-0.019	-0.001	-0.011	0.002	0.004	-0.003	0.015	-0.006	0.017	0.008	0.004	-0.013	0.001	-0.024
OFF BOXOUTS	-0.073	0.019	-0.043	-0.034	0.090	-0.039	0.011	-0.008	-0.015	0.023	-0.029	-0.030	0.125	0.043
DEF BOXOUTS	-0.158	0.010	-0.075	-0.126	0.366	-0.155	0.075	-0.008	0.011	0.075	-0.067	-0.050	0.194	0.209

Table 9: Clusters' centroid position in Principal Component space

Cluster	PC 1	PC 2	PC 3	PC 4	PC 5	PC 6	PC 7	PC 8	PC 9	PC 10	PC 11	PC 12	PC 13	PC 14
1	-0.06	0.05	0.32	0.18	0.08	0.03	-0.01	-0.01	-0.01	0.02	0.01	0.01	-0.02	0.00
2	0.16	-0.28	0.28	-0.25	-0.07	-0.06	0.03	0.01	-0.02	-0.01	-0.00	0.01	-0.01	0.02
3	0.72	0.20	-0.24	-0.20	-0.02	0.02	-0.00	0.02	0.06	0.02	0.01	0.00	0.01	-0.02
4	-0.78	0.17	-0.14	-0.10	0.16	-0.03	-0.02	0.13	0.00	0.03	0.02	-0.03	0.06	-0.03
5	0.49	0.61	0.01	0.13	0.06	0.11	-0.05	-0.06	-0.05	-0.02	0.00	-0.02	0.01	-0.02
6	0.08	-0.51	-0.10	0.05	-0.00	-0.12	-0.21	-0.08	0.13	0.04	-0.01	0.02	-0.03	0.03
7	0.45	-0.69	-0.43	0.28	0.09	-0.01	0.14	0.06	-0.17	-0.05	-0.01	-0.02	-0.00	-0.01
8	-0.48	-0.00	-0.05	0.16	-0.50	0.16	0.03	0.09	0.08	0.08	-0.02	-0.01	0.01	0.00
9	-0.78	-0.24	-0.34	-0.16	0.12	0.17	0.02	-0.22	0.02	-0.11	-0.05	-0.04	0.09	-0.03
10	-0.59	0.61	-0.21	0.14	-0.18	-0.31	0.17	-0.07	-0.02	-0.09	-0.01	0.09	-0.15	0.10

Table 10: Clusters' centroid position in feature space

cluster	1	2	3	4	5	6	7	8	9	10
EFG PCT	0.013	0.004	-0.013	0.052	0.004	-0.030	-0.094	-0.003	-0.008	0.013
USG PCT	-0.006	-0.054	0.047	-0.039	0.138	-0.063	-0.047	-0.031	-0.052	0.177
MIN	0.101	-0.018	0.020	-0.028	0.248	-0.181	-0.282	-0.082	-0.262	0.220
PTS OFF TOV	0.004	-0.003	0.009	-0.009	0.027	-0.013	-0.018	-0.015	-0.016	0.017
PTS 2ND CHANCE	-0.010	-0.036	-0.060	0.096	-0.019	-0.029	-0.087	0.060	0.111	0.100
PTS FB	0.005	0.006	0.014	-0.019	0.029	-0.008	-0.015	-0.016	-0.024	-0.012
PTS PAINT	-0.030	-0.065	-0.017	0.110	0.049	-0.066	-0.117	0.054	0.092	0.132
DRIVE PTS	-0.009	-0.019	0.109	-0.064	0.148	-0.039	-0.015	-0.083	-0.118	-0.010
DRIVE FG PCT	0.024	0.003	0.032	0.103	0.057	-0.067	-0.135	-0.005	-0.322	0.074
CATCH SHOOT PTS	0.050	0.046	-0.025	-0.055	-0.014	0.011	-0.012	-0.058	-0.090	0.025
CATCH SHOOT FG PCT	0.041	0.016	0.008	0.004	0.036	-0.050	-0.120	-0.016	-0.092	0.026
PULL UP PTS	0.006	-0.006	0.096	-0.088	0.123	-0.030	0.001	-0.088	-0.125	0.001
PULL UP FG PCT	0.039	0.013	0.056	-0.015	0.058	-0.053	-0.092	-0.026	-0.230	0.050
PAINT TOUCH PTS	-0.032	-0.068	-0.105	0.194	-0.064	-0.038	-0.127	0.126	0.217	0.152
PAINT TOUCH FG PCT	0.039	0.063	-0.010	0.057	0.028	0.029	-0.528	0.014	0.042	0.063
POST TOUCH PTS	0.006	-0.066	-0.083	0.055	0.003	-0.054	-0.051	0.094	0.040	0.293
POST TOUCH FG PCT	0.273	-0.269	-0.276	0.177	0.269	-0.249	-0.258	0.083	0.007	0.177
ELBOW TOUCH PTS	-0.014	-0.021	-0.044	0.090	-0.024	-0.064	-0.071	0.072	0.075	0.102
ELBOW TOUCH FG PCT	0.065	0.180	0.012	0.042	0.108	-0.387	-0.475	0.003	-0.059	0.045
FG2A	-0.024	-0.061	0.009	0.039	0.071	-0.057	-0.058	0.043	0.034	0.141
FG2 PCT	0.005	-0.000	-0.025	0.109	-0.002	-0.044	-0.181	0.045	0.016	0.020
FG3A	0.069	0.089	0.044	-0.188	0.050	0.060	0.069	-0.233	-0.186	-0.057
FG3 PCT	0.066	0.053	0.052	-0.090	0.064	0.006	-0.072	-0.129	-0.208	-0.035
OFF LOSSES RECOV.	-0.009	-0.001	0.007	0.053	0.011	-0.002	-0.021	-0.117	0.037	0.006
DEF LOSSES RECOV.	-0.004	0.005	0.011	0.006	0.006	0.002	0.008	-0.061	0.008	-0.003
PFD	-0.017	-0.051	0.012	0.037	0.068	-0.051	-0.052	0.010	0.018	0.116
DRIVE PF	-0.006	-0.016	0.067	-0.053	0.102	-0.022	-0.003	-0.048	-0.063	-0.005
AVG SPEED OFF	-0.010	0.021	0.061	-0.023	0.013	0.024	0.007	-0.045	-0.011	-0.147
AVG SPEED DEF	-0.006	0.025	0.002	-0.004	-0.033	0.024	0.023	0.001	0.018	-0.085
STL	-0.007	-0.001	0.028	-0.023	0.028	0.001	0.006	-0.034	-0.012	0.006
BLK	-0.009	-0.032	-0.059	0.096	-0.038	-0.016	-0.054	0.053	0.096	0.078
DEF RIM FGM	-0.011	-0.020	-0.037	0.067	-0.035	-0.005	-0.021	0.021	0.071	0.051
DEF RIM FGA	-0.012	-0.026	-0.048	0.083	-0.043	-0.009	-0.026	0.034	0.083	0.067
DEF RIM FG PCT	0.006	0.031	0.060	-0.058	0.044	0.012	-0.039	-0.059	-0.071	-0.055
D FGM	-0.023	-0.018	-0.023	0.045	-0.044	0.017	0.028	0.015	0.090	0.020
D FGA	-0.022	-0.017	-0.025	0.045	-0.046	0.018	0.031	0.015	0.088	0.020
D FG PCT	-0.007	-0.005	-0.003	0.010	-0.008	0.003	0.004	0.004	0.023	0.008
OPP PTS FB	-0.003	-0.002	-0.000	0.001	0.003	-0.009	-0.002	0.012	0.005	0.006
OPP PTS PAINT	-0.002	0.003	0.004	0.007	0.005	0.001	0.001	-0.038	0.010	-0.007
DEFLECTIONS	-0.005	0.011	0.037	0.021	0.037	0.006	-0.003	-0.228	0.008	0.025
CHARGES DRAWN	0.003	-0.003	0.008	0.009	0.018	-0.011	-0.009	-0.056	-0.013	0.026
CONTESTED SHOTS 2PT	-0.028	-0.051	-0.097	0.264	-0.089	-0.028	-0.065	-0.163	0.259	0.126
CONTESTED SHOTS 3PT	0.006	0.015	-0.009	0.029	-0.011	0.020	0.021	-0.159	0.036	-0.008
SCREEN ASSISTS	-0.035	-0.062	-0.072	0.230	-0.069	-0.048	-0.065	-0.043	0.217	0.108
SCREEN AST PTS	-0.039	-0.068	-0.080	0.254	-0.077	-0.053	-0.070	-0.050	0.240	0.116
PAINT TOUCH PASSES	-0.022	-0.042	-0.049	0.105	-0.045	-0.023	-0.051	0.101	0.101	0.075
PAINT TOUCH AST	-0.006	-0.045	-0.028	0.087	0.009	-0.050	-0.084	0.052	0.026	0.131
PAINT TOUCHES	-0.022	-0.046	-0.064	0.116	-0.043	-0.024	-0.056	0.089	0.128	0.091
POST TOUCH PASSES	-0.009	-0.045	-0.049	0.044	-0.013	-0.039	-0.036	0.080	0.023	0.237
POST TOUCH AST	-0.011	-0.059	-0.056	0.047	-0.009	-0.050	-0.060	0.105	0.003	0.346
POST TOUCHES	-0.008	-0.065	-0.080	0.058	-0.014	-0.054	-0.047	0.115	0.043	0.330
ELBOW TOUCH PASSES	-0.034	-0.064	-0.067	0.145	-0.067	-0.055	-0.073	0.167	0.100	0.200
ELBOW TOUCH AST	-0.023	-0.039	-0.029	0.086	-0.032	-0.037	-0.051	0.080	0.054	0.138
ELBOW TOUCHES	-0.043	-0.085	-0.101	0.204	-0.081	-0.086	-0.108	0.210	0.155	0.268
AST	-0.054	-0.073	0.219	-0.069	0.207	-0.074	-0.015	-0.097	-0.121	0.053
FT AST	-0.007	-0.010	0.026	-0.017	0.027	-0.009	0.003	0.001	-0.013	0.003
SECONDARY AST	-0.008	-0.009	0.069	-0.038	0.068	-0.022	-0.011	-0.038	-0.052	-0.006
POTENTIAL AST	-0.063	-0.083	0.249	-0.094	0.233	-0.080	-0.003	-0.080	-0.130	0.042
PASSES MADE	-0.073	-0.108	0.190	0.020	0.151	-0.086	-0.045	-0.014	-0.033	0.090
PASSES RECEIVED	-0.070	-0.089	0.274	-0.105	0.255	-0.091	0.005	-0.094	-0.145	0.052
DRIVE PASSES	-0.022	-0.014	0.113	-0.063	0.105	-0.020	0.012	-0.068	-0.067	-0.043
DRIVE AST	-0.012	-0.010	0.064	-0.033	0.059	-0.011	0.006	-0.034	-0.041	-0.021
TOUCHES	-0.069	-0.116	0.194	-0.002	0.191	-0.098	-0.052	-0.031	-0.050	0.137
AVG DRIB PER TOUCH	-0.072	-0.047	0.343	-0.203	0.302	-0.054	0.096	-0.176	-0.200	-0.154
TOV	-0.031	-0.047	0.056	-0.004	0.073	-0.033	-0.003	-0.007	0.008	0.072
OREB UNCONTEST	-0.014	-0.016	-0.017	0.050	-0.019	-0.013	-0.007	0.024	0.057	0.014
OREB CONTEST PCT	0.037	-0.075	-0.175	0.187	-0.045	0.035	-0.340	0.206	0.164	0.185
OREB CHANCE DEFER	0.005	-0.018	-0.006	0.028	0.031	-0.035	-0.051	0.011	0.001	0.041
DREB CONTEST	-0.011	-0.033	-0.055	0.081	-0.039	-0.017	-0.044	0.068	0.082	0.104
DREB UNCONTEST	-0.009	-0.039	-0.036	0.074	-0.006	-0.021	-0.049	0.029	0.067	0.095
DREB CONTEST PCT	-0.001	-0.031	-0.096	0.102	-0.075	-0.013	-0.061	0.115	0.094	0.128
DREB CHANCE DEFER	-0.002	-0.007	-0.010	0.017	-0.010	-0.000	-0.003	0.009	0.021	0.011
OFF BOXOUTS	-0.011	-0.026	-0.034	0.091	-0.025	-0.015	-0.026	-0.018	0.087	0.038
DEF BOXOUTS	-0.018	-0.034	-0.080	0.219	-0.088	-0.011	-0.044	-0.138	0.195	0.074